参考元素周期表，将以下几组原子和分子的性质排序，并做简单解释

(a)Ar，Ca^{2+}，Cl^{-}，Fe^{2+}: 原子或离子半径由小到大

(b)Cl，H，Mg，S: 电负性由小到大

(c)CO_2，CrO_3，Mn_2O_7，P_2O_5: 在水中酸性从弱到强(提示:考虑主副族对应关系).

(a) Ar Ca^{2+} Cl^- Fe^{2+}

Ar (+18) 2, 8, 8

Ca^{2+} (+20) 2, 8, 8

Cl^- (+17) 2, 8, 8

$Fe^{2+} < Ca^{2+} < Ar < Cl^-$

(b) Cl, H, Mg, S

$Mg < H < S < Cl$

(c) CO_2, Cr_2O_3, Mn_2O_7, P_2O_5

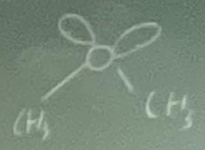
$+H_2O$ H_2CO_3 $H_2Cr_2O_4$ $H_2Mn_2O_8$ $H_2P_2O_6$

所在行 C S [illegible] P

$CO_2 < P_2O_5 < Cr_2O_3 < Mn_2O_7$

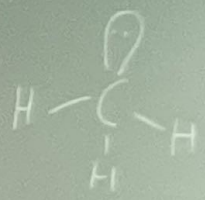
(a) CH_3-O

(a) $CH_3-\ddot{O}-CH_3$



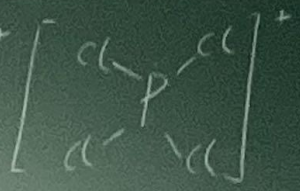
V形

(b) CH_3^-



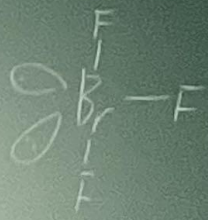
三角锥

(c) PCl_4^+



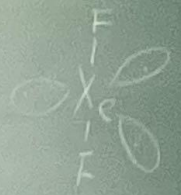
四面体

(d) BrF_3



T形

(e) XeF_2



直线形

b) Cl, H, Mg, S

中性的氧至少有O、O_2、O_3三种存在形式，每种原子或分子形式又可能形成一价或二价负离子：O^{n-}、O_2^{n-}、O_3^{n-}（n = 1或2）。

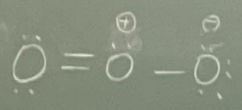
(a) 推导下面四种负氧离子中O—O键的平均键级. 要求写出过程和结果，可以任选分子轨道、Lewis结构式等方法.

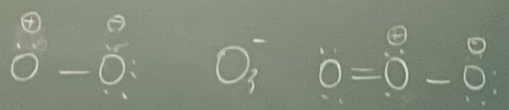
$$O_2^- \qquad O_2^{2-} \qquad O_3^- \qquad O_3^{2-}$$

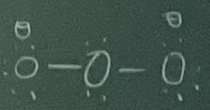
(b) 如果认为未成键离子的键级为零，根据键级大小与稳定性的关系，以上四种负氧离子中最稳定的是哪一种

(a) O_2^- $6+6+1=13$ $(2\sigma_{2s})^2(2\sigma_{2s}^*)^2(2\sigma_{2p_x})^2(\pi_{2p_y})^2(\pi_{2p_z})^2(\pi_{2p_y}^*)^2(\pi_{2p_z}^*)^1$ $\frac{3}{2}$

O_2^{2-} $6+6+2=14$ $(2\sigma_{2s})^2(2\sigma_{2s}^*)^2(2\sigma_{2p_x})^2(\pi_{2p_y})^2(\pi_{2p_z})^2(\pi_{2p_y}^*)^2(\pi_{2p_z}^*)^2$ 1

O_3 

O_3^- 

O_3^{2-} 

键级 σ $1\times2=2$

π $1=1$

平均键级 $\frac{2+1}{2}=\frac{3}{2}$

键级 σ $1\times2=2$

π 0.5

平均键级 $\frac{2+0.5}{2}=1.25$

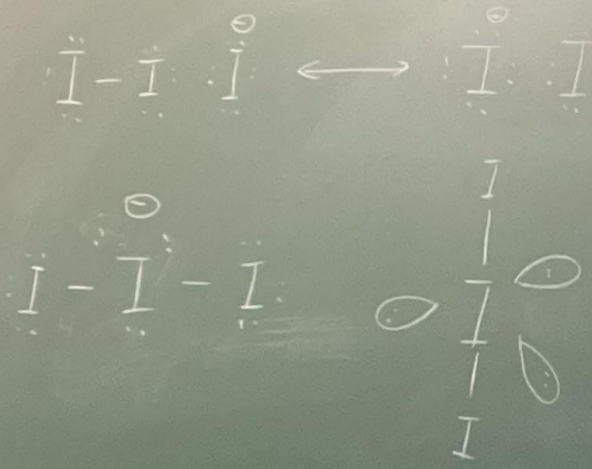
键级 σ $1\times2=2$

π 0

平均键级 $\frac{2+0}{2}=1$

O_3 O_3^- O_3^{2-}

(a) I_3^- $I_3^- \rightleftharpoons I_2 + I^-$



写出下列分子和离子的Lewis结构式。要求:

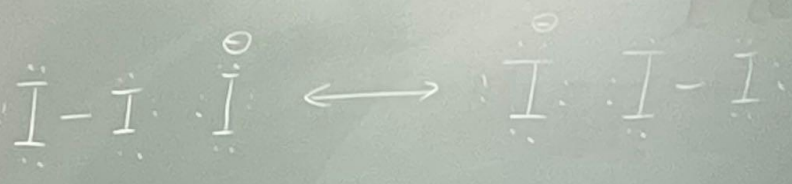
- 所有非氢原子满足八隅体规则并直线连接
- 用直线表示一对共享电子不能省略孤对电子, 不要使用虚线或离域π键来标记
- 实际电荷或形式电荷标记在原子旁边, 而不要给整个离子标记电荷
- 如果有数量提示, 则写出能量最低的几个共振结构

(a) I_3^- (2个共振式)

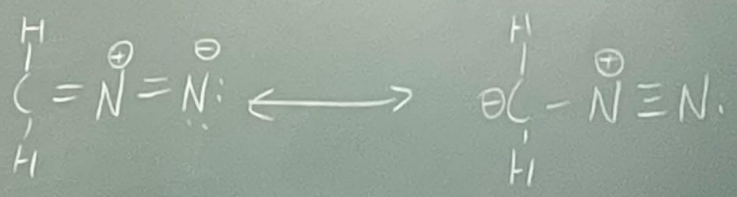
(b) CH_2N_2 (2个共振式)

(c) 写出C_3H_4的2种可能结构, 其中3个C均为直线连接但C与H的连接方式不同

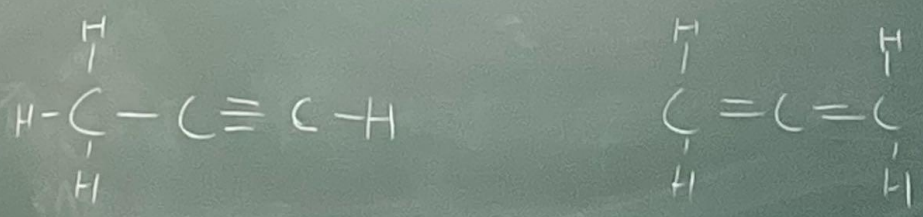
(a) I_3^- $\qquad I_3^- \rightleftharpoons I_2 + I^-$



(b)



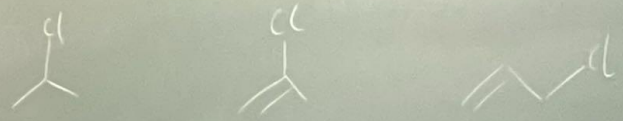
(c)



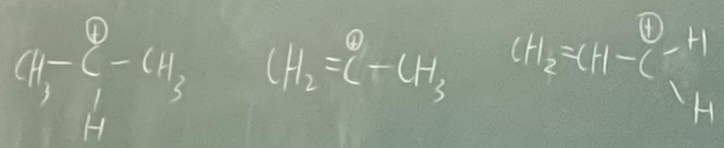
根据2-氯丙烷、2-氯丙烯、3-氯丙烯分子中的C—Cl键能，回答以下问题，填入表格的对应位置中

分子			
C—Cl键能($kJ\cdot mol^{-1}$)	305	389	251
C—Cl键上碳原子杂化类型			
失去Cl-后的碳正离子结构(画出所有的H)			
以上碳正离子的形状(描述C-C-C键角)			
以上碳正离子如有共振结构请画出			

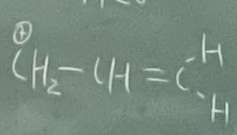
(c)



sp^3 sp^2 sp^3



约120° 180° 约120°



sp	$-C\equiv$	$-\overset{\oplus}{C}=$
sp^2	$-\overset{\vert}{C}=$	$-\overset{\vert}{C}-$ ⊕
sp^3	$-\overset{\vert}{C}-$	$-\overset{\vert}{C}-$ ⊖